From scaling laws to forecasts: how aggregation determines power-law verdicts and conditions predictive skill in Brazilian dengue surveillance

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Abstract

Power-law scaling is routinely reported for infectious disease incidence, and dengue in Brazil is a standard example: heavy-tailed distributions of case counts are invoked to argue that extreme epidemics are structurally unpredictable, and therefore that anticipating them is a matter of luck rather than skill. We test both halves of that argument, on the same data, at once. Using weekly dengue surveillance data spanning January 2010 to January 2026 for the 5,571 Brazilian municipalities, processed through the Mosqlimate/Infodengue platform, we fitted discrete power laws by maximum likelihood with Kolmogorov–Smirnov $x_{\min}$ selection at seven nested aggregation levels, compared each fit against an exponential alternative by a likelihood-ratio test, and then asked whether the resulting scaling descriptors add out-of-sample predictive value beyond lagged case counts, demographics and climate.

The descriptive results split sharply. For the cross-sectional distribution of *epidemic size* across municipalities within a year, the power law is preferred in 100% of the 15 national fits (all significant at $p < 0.05$; $\alpha \approx 2.0$) and in 96% of 330 state-year fits, with $x_{\min}$ in the hundreds of cases. For the distribution of *epidemic duration*, the power law is rejected in all 15 national fits (and in 95% of the state-year fits) in favour of an exponential, with α piling up at 2.90–2.96. For weekly case counts, the power law appears preferred in 75% of city-level series but $x_{\min} \leq 5$ in 77% of them, meaning that the fitted object is the bulk of the distribution rather than a tail.

The predictive half turns the description into a testable one. A power law with $\alpha \approx 2$ fixes the whole exceedance curve, so the fitted pair $(\alpha, x_{\min})$ is a two-parameter statement about how often extreme seasons occur — and, unlike the case counts themselves, the exponent is empirically stable relative to their two-order-of-magnitude variation, even though it is not independent of the scale of the season: within units, α and $x_{\min}$ are strongly rank-correlated, so a scaling block may be re-encoding the lagged case count rather than adding to it. We therefore use the lagged scaling descriptors as features in (i) random-forest and gradient-boosted quantile regressions of epidemic size, duration and peak timing, and (ii) a classifier of whether a municipality will exceed its historical 75th percentile. Because $\alpha \approx 2$ makes squared-error losses dominated by a few tail events, evaluation is on proper scoring rules — pinball loss and the Brier score — over a rolling-origin design with five forecast origins, and a four-rung ablation ladder isolates the incremental contribution of the scaling block from that of lagged metrics, population and climate.

The predictive half returns a negative result, and a sharp one. On a rolling-origin design with five forecast origins at state level, and a decision rule fixed in advance,

the scaling block never improves on the predictors it is added to: it gains on the preceding rung with a 95% bootstrap interval excluding zero in 0 of 5 origins for epidemic size, 0 of 5 for duration and 1 of 5 for peak week, against the 3 of 5 required. It is also worst where it was expected to be best: at the 95th percentile of the size forecast it *worsens* the pinball loss in all five origins, by 0.2% to 25% (median 18%), consistent with $\hat{\alpha}$ and $\hat{x}_{\min}$ acting as a noisier duplicate of the lagged case count. No rung beats calibrated persistence for size, whereas duration is genuinely forecastable (median skill 0.27); an apparent R^2 of 0.35 for size, reported by an earlier run of the same pipeline, was computed on the pooled cross-section on the log scale against no benchmark, and reflected between-unit differences that persistence reproduces trivially.

We conclude that the heavy-tail literature on dengue conflates at least three distinct random variables, that scale claims are only interpretable where $x_{\min}$ lies far above the bulk, and that reporting α without the (unit, $x_{\min}$, comparison model) tuple is uninformative — descriptive or predictive. A heavy tail is not, by itself, evidence that epidemics cannot be anticipated: it bounds how well the largest event's magnitude can be estimated from second moments, and leaves the probability of exceeding any level fully specified.

1 Introduction

Understanding the dynamics of infectious disease epidemics has led to a very prolific literature, both in terms of mechanistic mathematical models and of statistical models (Siettos and Russo, 2013). The former attempt to capture the causal chain of events involved in disease transmission, the latter try to make sense of a growing collection of data about infectious disease incidence.

Among the most salient results of the mathematical modelling of infectious diseases was the definition of the basic reproductive number, R_0, which determines the most important characteristics of epidemics such as their spreading speed and the epidemic threshold, also known as the herd immunity threshold. R_0 appears in the equations governing prevalence growth as an exponent determining the initial exponential growth of an epidemic, and it defines an important bifurcation point at $R_0 = 1$ linked to the existence and stability of equilibria in epidemic models.

On the statistical side, the largest portion of the literature deals with statistical modelling of incidence data, i.e. the number of cases accumulated over time and space or studied as a stochastic process, such as a Poisson or negative binomial process (Becker and Britton, 1999; Li et al., 2012). The central difficulty is overdispersion: real surveillance counts produce far more extreme observations than any tractable count model comfortably allows.

A recurrent claim is that this overdispersion is not merely large but *scale-free*, i.e. that incidence follows a power law (Rhodes et al., 1997; Clauset et al., 2009; Neipel et al., 2020). The attraction of this claim is that the very heavy tail implies extreme values that most models fail to predict — and those are precisely the events that cause the most disease and the largest economic burden. If true, this would have direct consequences for forecasting and for resource allocation.

The problem is that the empirical record is inconsistent. Reported exponents for dengue range widely, some studies find exponential rather than power-law behaviour, and the presence of a heavy tail appears to depend on the study period, the spatial unit and the outcome variable. Previous work has usually attributed this heterogeneity to biology or geography: different places, or different years, are assumed to be governed by different

regimes.

Here we test a simpler and more uncomfortable explanation: that the heterogeneity is largely produced by the analyst. Any epidemiological time series can be cut into distributions in several legitimate ways — the weekly counts of one city, the annual totals of one city, the set of epidemic sizes across cities in one year, the set of epidemic durations — and these are different random variables with different supports and different tail behaviour. If the verdict changes as the construction changes, then a substantial part of the published disagreement is methodological rather than biological.

Why the description matters for forecasting. The dispute would be of limited practical consequence if it stayed on the descriptive side. It does not. The heavy-tail claim is routinely used as an argument about forecastability: if epidemic sizes are drawn from a power law with exponent near 2, then the variance of the size distribution is dominated by its extreme upper tail, every squared-error forecast metric is decided by a handful of observations, and the honest conclusion is that point forecasts of epidemic size are unreliable. That inference is correct as far as it goes, but it says nothing about whether *probabilistic* forecasts are possible. A power law is a complete generative statement: it assigns a probability to every exceedance level, so the pair $(\alpha, x_{\min})$ summarises the whole risk curve of a municipality, and a stable α is a low-variance quantity in a setting where the observables are wildly volatile. Whether that summary carries information not already present in the lagged case counts is an empirical question, and it is the question the second half of this paper puts to the test: we first establish the scaling description, then submit it to an out-of-sample predictive evaluation in which it can either earn its place among the predictors or be shown to be redundant.

1.1 Contributions

This paper makes four contributions.

1. **A systematic seven-level comparison on one dataset.** We fit the same estimator to the same underlying surveillance data under seven nested aggregation levels (Table 1), so that any difference in the inferred regime is attributable to the construction and not to differing data sources, periods or software.

2. **An identification diagnostic that separates signal from artefact.** We show that the reported preference for a power law is only interpretable when the estimated lower bound $x_{\min}$ lies well above the bulk of the data. Where $x_{\min}$ collapses to 1–5 cases, the fit describes the entire distribution and the "tail" language is vacuous. We report $x_{\min}$ alongside every exponent and use it to classify each of the 47,494 fits.

3. **A resolved claim about dengue.** Epidemic *size* is heavy-tailed with $\alpha \approx 2$ and is stable across states and years; epidemic *duration* is not heavy-tailed and is better described by an exponential. The two variables are frequently conflated in the literature, which explains a large part of the apparent contradiction.

4. **A predictive test of the scaling description, and its answer.** We do not stop at the exponent. Treating $(\alpha, x_{\min})$ and the historical trend of α as lagged features, we fit gradient-boosted quantile regression models for epidemic size, duration and peak timing under a rolling-origin design with five forecast origins, and use a four-rung ablation ladder to measure whether the scaling block adds out-of-sample skill

over lagged counts, population and climate. It does not: the block fails the pre-registered rule on all three targets, and at the upper quantile of the size forecast it is measurably harmful, which is what the measured coupling between α and $x_{\min}$ predicts. The ablation converts a descriptive controversy into a falsifiable claim about forecast skill, and the claim is falsified.

2 Methods

2.1 Data sources

Weekly dengue case counts were obtained from the InfoDengue/Mosqlimate platform (Araujo et al., 2025) for the period from 1 January 2010 to 22 January 2026 (the extraction date), for all Brazilian municipalities with available data. The variable used is the nowcast-adjusted estimated number of cases (`casos_est`), resampled to epidemiological weeks ending on Sunday (`W-SUN`); zero-case weeks are retained in the series but excluded from distributional fits, since the power-law family is defined on positive support.

Annual epidemic descriptors were obtained from the EpiScanner module of the same platform, which partitions each municipal series into epidemic episodes and reports, for each episode, its total size (*total cases*) and its duration in weeks (*ep_dur*). Annual cross-sections are available for the 15 epidemic years 2011–2025; the pooled all-years fit is stored separately (with a sentinel `year = 0`) and is excluded from all year counts. This yields, for each state-year and for each national year, a cross-section of municipal epidemics.

Municipal population figures were taken from the estimates distributed with the platform (`city_stats` table; 5,571 municipalities, one with a zero entry excluded). Population enters no power-law fit; it is used to compute the descriptive correlations of Section 3.1 and, as $\log_{10}$ population, as one predictor in the forecasting models of Section 2.7.

2.2 Seven aggregation levels

The unit of analysis is the *set of positive observations* passed to the estimator. Table 1 defines the seven constructions used. The first three treat the individual municipality as the sampling unit and vary the temporal window; the last four treat a cross-section of municipalities within a period as the sampling unit.

Table 1: The seven aggregation levels. "Unit" is the sampling unit, i.e. what each observation in the fitted sample represents.

ID	Unit	Variable	Source table
L1	city	weekly cases, full series	`powerlaw_fits`
L2	city-year	weekly cases, one season	`powerlaw_fits_yearly`
L3	city	annual total, across years	`powerlaw_fits_total_cases`
L4	city $\subset$ state-year	epidemic size	`episcanner_state_fits`
L5	city $\subset$ state-year	epidemic duration	`episcanner_state_fits`
L6	city $\subset$ national-year	epidemic size	`episcanner_fits`
L7	city $\subset$ national-year	epidemic duration	`episcanner_fits`

Levels L1 and L2 differ only in the temporal window over which the weekly counts are pooled: L1 uses the complete available series for a municipality (January 2010 to January 2026), whereas L2 restricts to a single epidemic year (at most 52 weeks). Both sample sizes therefore differ by roughly an order of magnitude, which is why we compare shares

rather than raw counts. Levels L4–L7 differ only in the spatial scope of the cross-section, not in the variable being measured. This nesting is deliberate: it allows each contrast in Section 3 to be read as the effect of a single change in construction. Figure 1 shows the scheme.

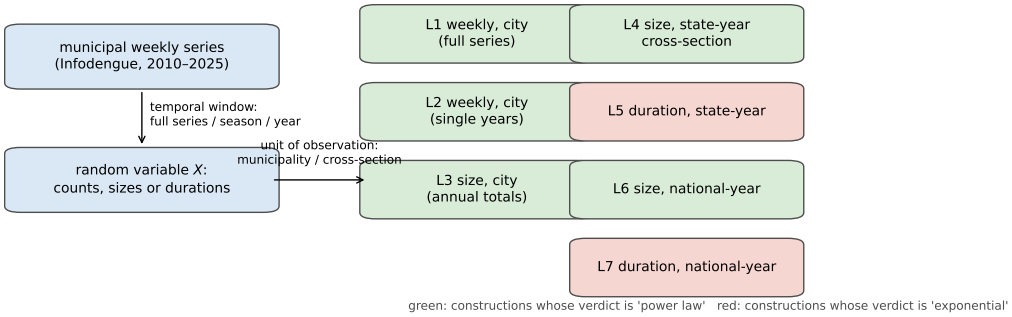


Figure 1: The seven aggregation levels as nested constructions on a single dataset. Starting from the same municipal weekly series (top), the analyst chooses a temporal window — the full series, one season, or one year — and a unit of observation — either the municipality itself, or the cross-section of municipalities within a period — and obtains random variables that differ in support, variance and tail behaviour. L1 and L2 differ only in the window; L3 aggregates each municipality over years; L4–L7 differ only in the spatial scope of the cross-section (state-year versus national-year). Every contrast reported in Section 3 is a single step in this scheme, so a difference in the inferred regime is attributable to the construction rather than to a change of data source, period or software.

2.3 Discrete power-law estimation

For each sample we fit a discrete power law by maximum likelihood following Clauset et al. (2009). The fitted density for integer $x \geq x_{\min}$ is

$$p(x) = \frac{x^{-\alpha}}{\zeta(\alpha, x_{\min})}, \tag{1}$$

where $\zeta(\alpha, x_{\min})$ is the generalised (Hurwitz) zeta function, which normalises the discrete mass. The exponent α is estimated by numerical maximisation of the likelihood, and the lower bound $x_{\min}$ is selected jointly by minimising the Kolmogorov–Smirnov distance between the empirical and fitted cumulative distributions over all candidate values of $x_{\min}$. Because the data are counts, the discrete formulation is used throughout (`discrete=True`); the continuous approximation is not employed at any level.

One property of the estimator has to be stated because it bears on the duration results. The numerical fit constrains α to $(1, 3]$: the upper limit of the optimiser's default parameter range is 3, so an estimate cannot exceed it and a sample whose likelihood is maximised beyond that value returns $\alpha = 3$ rather than its unconstrained maximum. The bound is immaterial where the fitted exponents lie well inside the range. At the cross-sectional size levels the largest value is 2.99 (L4) and 2.27 (L6); at the two weekly levels it is reached by 2 and 5 fits respectively, under 0.05% of each. It binds materially only for durations, where the upper quartile of the 330 state-year fits lies above 2.92 and

11 of them (3.3%) sit at the ceiling. Those fits should be read as *at least* 3, not as 3, and a genuinely scale-free body would not systematically exhaust the upper bound of the estimator.

The choice of $x_{\min}$ is not a nuisance parameter. It determines which fraction of the data the model actually describes, and therefore whether the fitted exponent refers to a tail at all. We therefore retain $x_{\min}$ for every fit and report its median and the proportion of fits with $x_{\min} \leq 5$ in each aggregation level.

2.4 Model comparison

A power law can always be fitted; the question is whether it describes the data better than a plausible alternative. We compare each power-law fit against an exponential distribution, which is the natural null for epidemiological counts because it corresponds to a constant hazard of termination and produces a light tail. The comparison follows the likelihood-ratio procedure of Clauset et al. (2009), as implemented in the `powerlaw` package with `normalized_ratio=True`.

$$R = \sum_{i=1}^{n} \left[\ln p_{\mathrm{PL}}(x_i) - \ln p_{\exp}(x_i) \right], \tag{2}$$

normalised so that its magnitude is comparable across samples of different size. The sign of R indicates the preferred model ($R > 0$ favours the power law, $R < 0$ favours the exponential) and its associated p-value quantifies whether the sign is distinguishable from sampling noise.

We adopt a deliberately conservative reporting convention. A level is described as *power-law supported* only when $R > 0$ and $p < 0.05$; as *exponential supported* only when $R < 0$ and $p < 0.05$; and as *inconclusive* otherwise. This prevents the common error of counting a positive but non-significant R as evidence for scaling.

2.5 Identification diagnostic

We flag a fit as *tail-informative* when $x_{\min}$ exceeds the bulk of the observed distribution, operationally $x_{\min} > 5$ cases for weekly counts and $x_{\min}$ above the median observed value for the aggregate levels. Where $x_{\min} \leq 5$, the fitted power law is being asked to describe the entire empirical distribution, including the small-count regime where the discrete power law is not a plausible generative model (case counts of 1 or 2 arise from reporting mechanics, not from scale-free branching). Such fits are retained in the database but are excluded from substantive interpretation, and we report the proportion explicitly.

2.6 From description to prediction

A fitted power law is not merely a description of a histogram; it is a statement about exceedance probabilities. For integer $s \geq x_{\min}$,

$$\Pr(S \geq s) \;=\; \frac{\zeta(\alpha, s)}{\zeta(\alpha, x_{\min})}, \tag{3}$$

so a single pair $(\alpha, x_{\min})$ parameterises the probability that a municipality experiences an epidemic of at least any given size. Four consequences of Equation (3) shape the forecasting design.

First, the exponent summarises *risk* rather than *activity*. Two municipalities with the same expected annual incidence but different α face different probabilities of a catastrophic season, and that difference is invisible to the mean. The fitted models quantify the claim rather than merely asserting it: using the 15 national size fits, they imply that between 5.4% and 16.5% of municipalities exceed ten times $x_{\min}$ in a given year, and between 0.3% and 2.7% exceed one hundred times $x_{\min}$. Those are falsifiable implications of the description, and Figure 9 confronts them with the empirical exceedance frequencies.

Second, α is a low-variance quantity *relative to its target*. Across the national cross-sections it takes values between 1.78 and 2.27, and it is essentially unrelated to city size ($r = -0.065$). Annual case counts over the same period vary by two orders of magnitude. The dispersion of $\hat{\alpha}$ is not zero — it exceeds sampling noise by about a factor of five in standard deviation, and it co-varies with the selected $x_{\min}$ (Section 3.3) — but it is small along precisely the dimension in which the target is volatile. That is the specific reason to carry α into the predictive models rather than treating it as a descriptive curiosity, and the same coupling is the reason the ablation, not the correlation, has to decide the question.

Third, $\alpha \approx 2$ has a direct methodological consequence for evaluation. The second moment of a power law with $\alpha \leq 2$ does not exist in the continuous limit, and at $\alpha \approx 2$ the sample mean is unstable, so squared-error losses are decided by the few largest outbreaks. A model that is good everywhere except in the tail can therefore look excellent or terrible depending on which years fall in the test set. We consequently evaluate on quantile (pinball) losses and on the classification of tail events, and treat RMSE as a secondary, tail-dominated diagnostic.

Finally, the predictive hypothesis is stated in a form that can fail. If $(\alpha, x_{\min})$ carries information about the coming season beyond what lagged counts already encode, a model containing them should beat an otherwise identical model without them, out of sample. If instead the exponent is close to a deterministic function of the lower moments of the same series, the two models should be indistinguishable — and the descriptive result would then carry no forecasting content. Section 2.10 specifies the comparison that decides between these outcomes, and Section 2.11 lists the three ways in which such a comparison could be faked and how each is prevented.

2.7 Predictive targets and feature construction

The forecasting problem is posed at the level of an administrative unit (municipality or state) and a target year t: every predictor is measured in year $t-1$ or earlier, and the target is a property of the epidemic season of year t. Three targets are modelled, corresponding to the descriptors that surveillance services actually act on: epidemic size S_t (total cases in the annual episode), epidemic duration D_t (weeks), and peak timing P_t (week of the seasonal maximum).

Predictors fall into six blocks, listed in Table 2: lagged epidemic metrics; the scaling descriptors $(\alpha_{S,t-1}, x_{\min,S,t-1})$ from the power-law fit to the previous season; the historical trend $\dot{\alpha}_{t-1}$, defined as the ordinary least-squares slope of α_S over all years strictly before t and set to zero where fewer than two such years exist; the epidemiological driver $R_{0,t-1}$ reported by the EpiScanner decomposition; $\log_{10}$ population; and eight quarterly climate aggregates from year $t-1$, namely mean minimum temperature and mean minimum humidity in each of Q1–Q4. Units lacking the mandatory lags (no previous season) are dropped rather than imputed; the optional $\dot{\alpha}$ is set to zero when undefined.

Table 2: Predictor blocks and the sources from which they are computed. All predictors refer to year $t-1$ or earlier. The scaling block is the object of the ablation in Section 2.10.

Block	Variables	Source
Lagged metrics	S_{t-1}, D_{t-1}, P_{t-1}	`episcanner`
Scaling descriptors	$\alpha_{S,t-1}$, $x_{\min,S,t-1}$	`powerlaw_results.db`
Scaling trend	$\dot{\alpha}_{t-1}$	derived (OLS on α_S)
Epidemiological	$R_{0,t-1}$	`episcanner`
Demographic	$\log_{10}(\text{pop})$	`city_stats`
Climate	$\bar{T}_{Q1..Q4}$, $\bar{H}_{Q1..Q4}$	Infodengue (`tempmin`, `umidmin`)

2.8 Regression models

Epidemic size is modelled on the log scale, $\log(1 + S_t)$, because the target spans four orders of magnitude and a model fitted on the raw scale would in practice be fitted to the largest municipalities only; predictions are returned to the case scale with $\exp(\cdot) - 1$. That back-transformation is biased downwards by Jensen's inequality, so a multiplicative factor $c = \overline{S}_{\text{train}} / \overline{(\exp \hat{y} - 1)}_{\text{train}}$ is estimated on the training data and applied to the point predictions. The correction is a mean-matching device, not a distributional one, and it is deliberately not applied to the quantile models, whose losses are computed on the back-transformed quantiles themselves. Duration and peak week are modelled on their natural scale.

Three estimators are used. (i) A random forest of 150 bootstrapped trees (Breiman, 2001), which additionally provides an out-of-bag R^2 reported as an in-sample fit diagnostic. (ii) A LightGBM gradient-boosted model with quantile objective at level τ (Ke et al., 2017; Koenker and Bassett, 1978), with 300 trees, learning rate 0.05, 16 leaves and a minimum of 5 observations per leaf. (iii) A histogram gradient-boosting model, also with quantile loss, 300 iterations, learning rate 0.05 and maximum depth 5. The quantile models are fitted on the grid $\tau \in \{0.05, 0.10, \ldots, 0.95\}$, which yields a full predictive distribution per unit and year rather than a point forecast. All stochastic components use `random_state=42`.

2.9 Classification of large epidemics

The second predictive task is posed as a decision problem rather than an estimation problem: will unit i experience, in year t, a season larger than the p-th percentile of its own historical distribution? The target is

$$Y_{i,t} \;=\; \mathbb{I}\big[\, S_{i,t} > Q_p\big(\{S_{i,k}\}_{k<t}\big) \,\big]\,, \tag{4}$$

with $p = 0.75$ by default. The threshold is a *within-fold* quantity: it is recomputed from the training years of each fold and each split, never from the full sample, so that the label of a test observation cannot depend on information unavailable to the model. The same discipline governs the class-imbalance weight, set to the negative-to-positive ratio of the training fold.

Two gradient-boosted classifiers are used, XGBoost (Chen and Guestrin, 2016) and LightGBM, with imbalance handled by `scale_pos_weight` and `is_unbalance` respectively. Validation is an expanding-window split defined on the year index. Fold j trains on every unit-year whose year is earlier than v_j and validates on year v_j alone, where the v_j are the $k = \min\{5, n_{\text{years}} - 2\}$ most recent training years: a fold exists only once at

least two earlier years are available, and the number of folds is capped at five. Because the folds are a function of the year vector and not of row position, they are invariant to the order of the rows in the frame. Within each fold the percentile threshold that defines the label, the class-imbalance weight and the median imputer are all recomputed from that fold's training years, so no quantity used to label or to weight a validation year can have been influenced by it; a per-fold audit table (training window, held-out year, fold sizes, threshold, weight) is retained with the results.

This distinction is not cosmetic, and it is the one step of this pipeline whose failure is silent. An implementation that applies a contiguous-index split to a frame sorted by unit rather than by year produces a grouped-by-municipality split: each fold trains on some municipalities over the whole calendar and validates on others over the same calendar, so validation contains no year that is absent from training. On the training panel used here — 32,824 rows, 4,725 municipalities, 2010–2023 — all five folds of such a split train on 2010–2023 and validate on 2010–2023, and 100% of validation rows carry a year already present in training. The quantity being estimated is then transportability to new municipalities, not temporal generalisation, and the estimate depends on an undeclared sort order: merely reordering the rows of a synthetic benchmark moved its cross-validated AUC from 0.813 to 0.835, whereas the year-indexed split returns the same folds and the same AUC under every permutation of the rows. The trap is narrower but not absent when the frame happens to be year-sorted, because contiguous blocks then still cut through calendar years. The splitter therefore takes the year vector explicitly, and `assert_forward_in_time` raises rather than warns when any validation year is not strictly later than the training years of its fold. The decision threshold applied to the predicted probability is chosen by Youden's J statistic (Youden, 1950), computed on the out-of-fold predictions rather than on the training fit, and then held fixed for the test set. We report the area under the ROC curve, the area under the precision–recall curve (the more informative summary under class imbalance), the Brier score (Gneiting and Raftery, 2007), the F_1 score at the Youden threshold and a calibration curve, and we test differences in AUC between nested feature sets by the method of DeLong et al. (1988). Feature attributions are obtained from SHAP values (Lundberg and Lee, 2017).

2.10 Evaluation protocol and ablation ladder

A single temporal split cannot distinguish a feature that helps from a feature that happened to help once. We therefore adopt a rolling-origin design: for each forecast origin $y \in \{2021, \ldots, 2025\}$ the model is trained on all observations with year $< y$ and evaluated on year y, and the per-origin losses are pooled. This gives five honest out-of-sample years and allows skill to be read as a function of the history available to the model.

Within each origin, the incremental value of the scaling description is isolated by the ablation ladder of Table 3. Each rung adds one block of predictors to the previous one, so a contrast between consecutive rungs is attributable to the block that was added. The rung of substantive interest is M2 versus M3, which differ *only* in the presence of $(\alpha_{S,t-1}, x_{\min,S,t-1}, \dot{\alpha}_{t-1})$.

Operationally, M0 forecast values are scaled by the empirical τ-quantile of the within-unit year-on-year ratio, estimated on the training years, so that the same benchmark can be evaluated at every quantile level rather than only at the median. At $\tau = 0.5$ the scale factor is close to unity and M0 reduces to plain persistence; at the other levels this is the honest analogue of persistence, not a straw man.

M0 is deliberately naive, because a benchmark that is already sophisticated manufac-

Table 3: The ablation ladder. M0 is the benchmark that any claim of skill must beat; successive rungs add one predictor block so that consecutive contrasts are attributable to that block. The M2→M3 step isolates the contribution of the power-law scaling descriptors.

Rung	Name	Predictors
M0	Persistence	$\hat{S}_t = S_{t-1}$, $\hat{D}_t = D_{t-1}$; historical median for P_t; training base rate for the classifier
M1	Lagged metrics	M0 predictors + P_{t-1}, $R_{0,t-1}$, $\log_{10}(\text{pop})$
M2	+ climate	M1 + quarterly $\bar{T}$, $\bar{H}$ from year $t-1$
M3	+ scaling	M2 + $\alpha_{S,t-1}$, $x_{\min,S,t-1}$, $\dot{\alpha}_{t-1}$

tures skill for whatever is added to it. Skill is reported as

$$\text{SS} \;=\; 1 - \frac{L(\text{model})}{L(\text{M0})}, \tag{5}$$

on the primary loss of each task — the pinball loss at $\tau = 0.5$ for each of the three regression targets, and the Brier score for the classifier, both proper scoring rules (Gneiting and Raftery, 2007). Secondary losses are MAE and RMSE of the point forecasts, the mean absolute scaled error of Hyndman and Koehler (2006) to make errors comparable across units of very different size, and the pinball loss at each τ on the grid.

Because units differ by orders of magnitude in scale, all model-versus-model comparisons are made on paired loss differentials computed within the same unit and year; their uncertainty is estimated by a bootstrap that resamples *units* rather than observations, preserving the within-unit correlation across years, with 2,000 replicates. For the pooled comparison across origins we additionally report the Diebold–Mariano statistic (Diebold and Mariano, 1995) with a Newey–West correction, since the loss series are serially correlated by construction.

The decision rule is fixed in advance. The scaling block is judged to add predictive value only if M3 improves on M2 on the primary loss with a 95% bootstrap interval for the paired difference that excludes zero, in the same direction, for at least three of the five forecast origins. This is stricter than a single significant pooled test, and deliberately so: a pooled test can be carried by one anomalous season, which is exactly the failure mode that a heavy-tailed target invites.

2.11 Leakage controls

Four sources of leakage are structurally possible in this design, and each is closed explicitly.

1. **Contemporaneous scaling descriptors.** $\alpha_{S,t}$ is estimated from the weekly counts of season t, which are part of the target. Only $\alpha_{S,t-1}$ enters as a predictor, and $\dot{\alpha}_{t-1}$ is computed exclusively from years strictly before t.

2. **The classification threshold.** Q_p is computed inside each fold from training years only. Computing it once on the full sample would leak the distribution of the test years into the labels of the training years, inflating apparent skill by construction.

3. **Cross-sectional information.** The cross-section of municipalities within year t is an outcome of season t and is therefore never a predictor; the scaling descriptors always come from the previous season's cross-section.

4. **The validation split itself.** Folds are defined on the year index (Section 2.9), so no fold trains on a year at or after the year it validates on. This is the only one of the four channels that fails silently in a plausible implementation — every number still looks like a respectable cross-validation score — which is why it is asserted in code rather than merely documented.

2.12 Software and reproducibility

All fits are stored in a single SQLite database (`powerlaw_results.db`) committed to the project repository, together with the fitting code (`fitpl.py`) and the cached parquet extracts of every model result. Every number in this manuscript is derived from that database by the queries in Appendix A; no statistic is transcribed by hand. The full pipeline is therefore re-runnable, and a reader can confirm or contradict each figure with a single SQL statement.

3 Results

Table 4 summarises the 47,494 fits. The contrast between levels is the central result of the descriptive half of the paper, and Figure 2 shows it at the level of the estimator's output.

3.1 Scaling verdicts

Table 4: Power-law fits by aggregation level. α median with interquartile range; $x_{\min}$ median; $\mathrm{PL}_{\mathrm{pref}}$ = proportion with $R > 0$; $\mathrm{PL}_{\mathrm{sig}}$ = proportion with $R > 0$ and $p < 0.05$; $\mathrm{Exp}_{\mathrm{sig}}$ = proportion with $R < 0$ and $p < 0.05$. Shares are percentages.

Level	n	α	IQR	$x_{\min}$	$\mathrm{PL}_{\mathrm{pref}}$	$\mathrm{PL}_{\mathrm{sig}}$	$\mathrm{Exp}_{\mathrm{sig}}$
L1 weekly, city	5,244	1.95	1.72–2.29	2	75	45	4
L2 weekly, city-year	40,616	2.06	1.72–2.45	2	23	3	21
L3 annual total, city	944	2.09	1.80–2.46	341	51	4	4
L4 size, state-year	330	1.97	1.79–2.14	159	96	52	0
L5 duration, state-year	330	2.80	2.52–2.92	17	0	0	95
L6 size, national-year	15	2.00	1.92–2.11	762	100	100	0
L7 duration, national-year	15	2.95	2.90–2.96	18	0	0	100

Epidemic size scales. At the national cross-section (L6), all 15 annual fits favour the power law and all are significant, with $\alpha = 2.00$ (IQR 1.92–2.11) and $x_{\min}$ of 762 cases. The exponent is remarkably stable: the annual values range only from 1.78 to 2.27. At state level (L4) the picture is consistent but weaker in significance, as expected from smaller cross-sections: the power law is preferred in 96% of 330 fits, significant in 52%, and α has a median of 1.97 across the 26 states (state-level medians spanning 1.70–2.58).

Duration does not scale. At the same cross-sectional levels, the duration of the same municipal epidemics behaves in the opposite way. At national level (L7) the power law is rejected in 15 of 15 years, with the exponential significant in every one. The estimated exponent piles up at 2.95 (full range 2.88–2.99, interquartile range 2.90–2.96, SD 0.03),

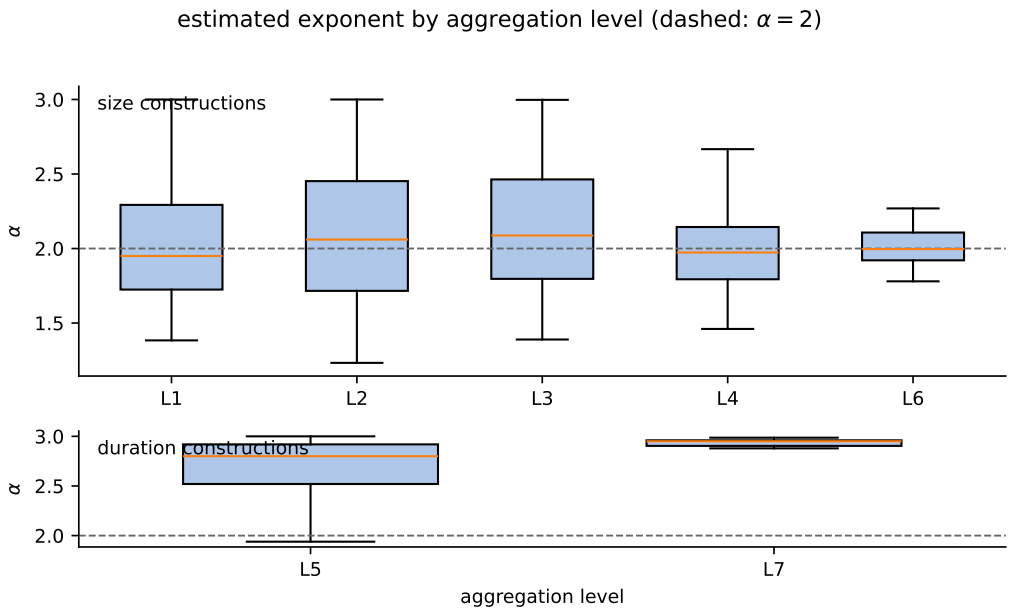


Figure 2: Distribution of the estimated exponent α at each aggregation level. Boxes give the median and interquartile range across fits, whiskers extend to 1.5 times the IQR, and the dashed line marks $\alpha = 2$. The upper panel groups the size constructions (L1–L4, L6) and the lower panel the duration constructions (L5, L7). Size distributions concentrate near $\alpha \approx 2$ at every level. Durations concentrate near $\alpha \approx 2.9$, which is close to the value at which a discrete power law approximates a truncated or exponential distribution, so the estimator returns a near-constant exponent regardless of the underlying process — a diagnostic that is easy to mistake for a physical result.

which is the signature of a truncated or exponential body being forced through a power-law estimator rather than evidence of scaling. The estimator's ceiling of $\alpha = 3$ (Section 2.3) is also in play: the upper quartile of the 330 state-year duration fits lies above 2.92 and 11 of them (3.3%) sit exactly at the bound, whereas no national size fit exceeds 2.27. At state level (L5) the exponential is significant in 95% of fits. Figure 3 summarises these shares across all levels, and Figure 4 shows the shape that produces them.

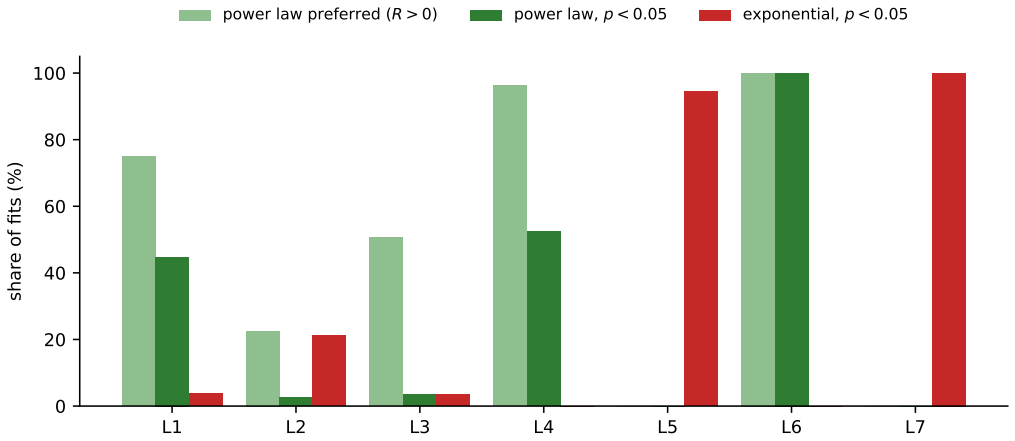


Figure 3: Model comparison by aggregation level. For each level, the proportion of fits in which the power law is preferred ($R > 0$), the proportion in which it is preferred *and* significant at $p < 0.05$, and the proportion in which the exponential is significant ($R < 0$ and $p < 0.05$). Levels are ordered from the individual-series constructions (left) to the cross-sectional ones (right). The two weekly levels produce apparent power-law support that the $x_{\min}$ diagnostic removes; the cross-sectional size levels are the only construction whose support survives both tests; the duration levels give the opposite verdict.

Weekly counts are not a tail. Levels L1 and L2 produce large apparent support for the power law (75% and, in the full-series case, 45% significant), but the estimated lower bound collapses to $x_{\min} = 2$ in the majority of fits: 77% of L1 and 81% of L2 fits have $x_{\min} \leq 5$. At that bound the model is fitted to essentially the whole distribution, including counts of 1 and 2 cases, and the heavy-tail interpretation does not follow. By contrast, in L3–L7 the median $x_{\min}$ is 17–762 cases, i.e. the fit describes genuine tails. Figure 5 plots the diagnostic directly, and Figure 6 shows what it means for individual fits.

The window flips the verdict on the same data. The cleanest demonstration is the L1/L2 contrast: pooling the same weekly counts over one season instead of a decade drops the proportion of fits favouring the power law from 75% to 23% and raises the proportion significantly favouring the exponential from 4% to 21%. Nothing about dengue changed; only the window did. Figure 7 shows the shift on the same underlying series.

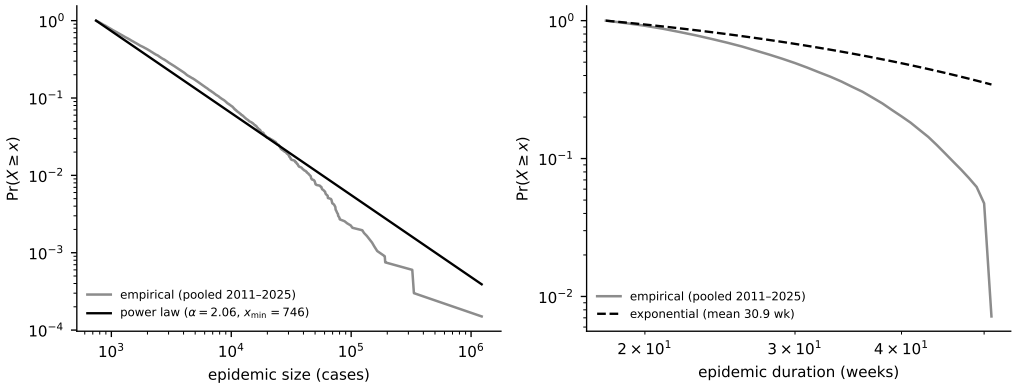


Figure 4: Empirical complementary cumulative distributions at the national cross-section, pooled over the 15 epidemic years, with the selected model superimposed. Left, epidemic size on log–log axes with the fitted power law: the curve is close to linear over two decades above $x_{\min}$. Right, epidemic duration on the same axes, where the curve bends downwards and is captured by the exponential. A straight line on the left and a convex curve on the right are the visual counterpart of the opposite verdicts in Table 4: the two variables are not two views of one process.

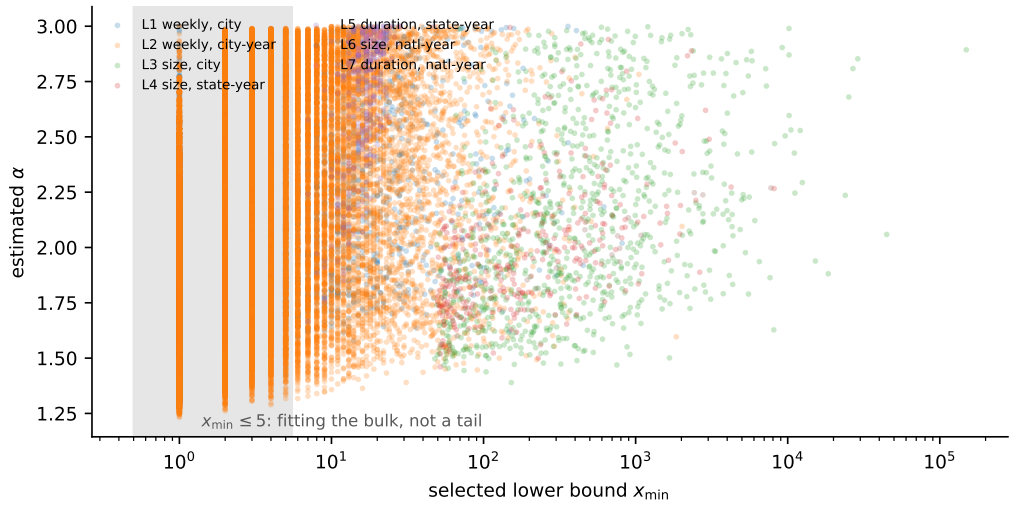


Figure 5: Identification diagnostic. Estimated exponent α against the selected lower bound $x_{\min}$ (log scale), one point per fit, coloured by aggregation level. The shaded band marks $x_{\min} \leq 5$, where the fitted power law is asked to describe the bulk of the distribution rather than a tail. The two weekly levels sit almost entirely inside that band, while the cross-sectional levels sit two to three orders of magnitude to its right. The dense vertical structure near $\alpha \approx 2.9$ for durations is the signature of a truncated distribution forced through a power-law estimator.

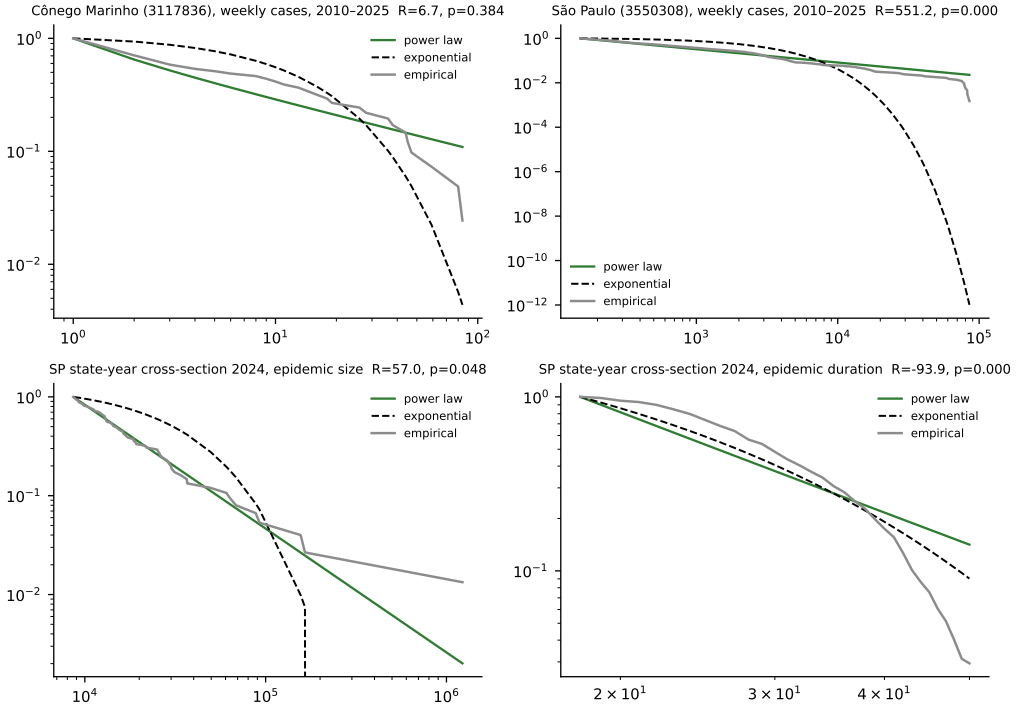


Figure 6: Individual fits across the aggregation levels, each with the power-law (solid) and exponential (dashed) alternatives superimposed. (a) Weekly counts of a small municipality, where the selected $x_{\min}$ collapses to 2 and the "power law" is fitted to the entire distribution; (b) weekly counts of a large state capital; (c) a state-year cross-section of epidemic sizes, where $x_{\min}$ lies in the hundreds of cases; (d) the corresponding cross-section of durations, where the exponential is preferred. Panels (a) and (c) are produced by the same estimator applied to different objects, and together they are the central distinction of the paper.

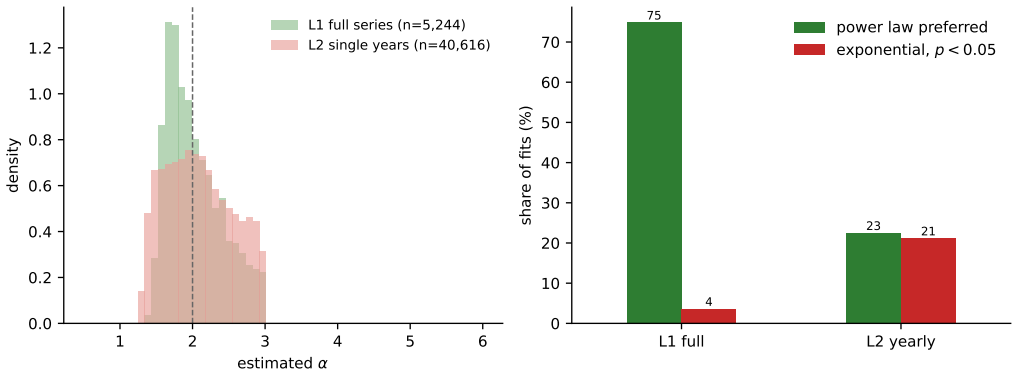


Figure 7: The window flips the verdict on identical data. Left, density of the estimated exponent from the full municipal series (L1) and from the same series cut into single seasons (L2). Right, the corresponding verdicts: the share of fits favouring the power law falls from 75% to 23%, while the share significantly favouring the exponential rises from 4% to 21%. No transmission parameter changes between the panels; only the length of the window does.

3.2 Descriptive associations

The municipal exponent is essentially unrelated to city size: $\mathrm{corr}(\alpha, \log_{10}\text{population}) = -0.065$ (Spearman -0.072; $n = 5{,}244$). The association with incidence is weak and negative (Pearson -0.136, Spearman -0.248; $n = 4{,}663$). We therefore find no support for the view that larger or more heavily burdened municipalities sit in a different scaling regime. The exponent is a property of the aggregation level and, within a unit, of the scale of the season (Section 3.3), not of the municipality. Figure 8 shows both the stability and the absence of association.

3.3 The exponent is coupled to the scale of the season

The fitted pair $(\alpha, x_{\min})$ is not a pair of independent coefficients. Table 5 collects the association between the two estimates at the levels where it can be measured, together with the association recovered from simulated data in which the true exponent was held fixed.

Table 5: Association between the estimated exponent and the selected lower bound. Rows 1–3 are the surveillance data; the last row is a control in which data were generated from a power law with true $\alpha = 2.0$ while the true lower bound was varied over the same range, so that any association recovered there is an artefact of the estimator. Values are Spearman rank correlations between α and $\log_{10} x_{\min}$.

Stratum	Sample	ρ	n
National cross-section, year to year	15 years	0.87	15
Within state, year to year	median of 24 states	0.65	–
Between states	state means	0.22	26
Simulated, α fixed at 2.0	control	-0.20	40

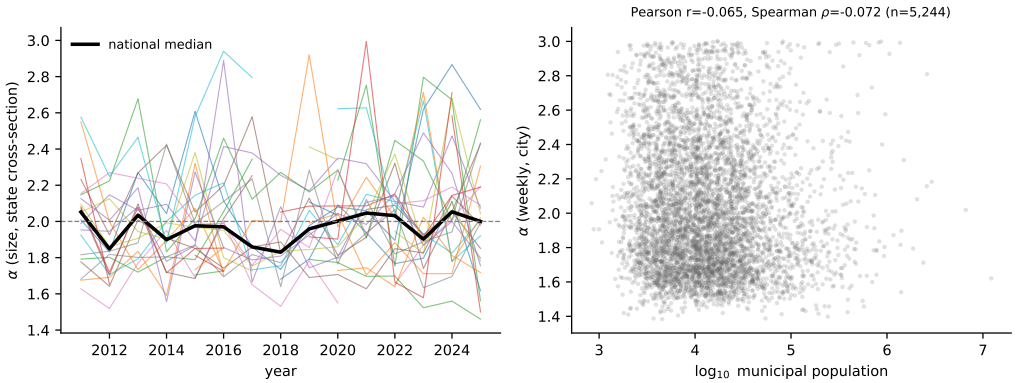


Figure 8: Stability of the exponent. (a) Estimated α for epidemic size by year, one line per state, at the state cross-section: the lines are flat, and the spread between states is far smaller than the year-to-year variation in case counts. (b) α against $\log_{10}$ municipal population, which shows no association (Pearson $r = -0.065$, Spearman -0.072, $n = 5{,}244$). Panel (a) is the empirical reason the exponent is a candidate predictor: it varies slowly along exactly the axis in which the observables do not.

Three features of the table matter. The surveillance association is strong at national level and strong *within* states over time — positive in 88% of the 24 states with at least six years of data, and significantly positive in 67% — but weak between states. The coupling is therefore a within-unit, over-time relationship rather than a contrast between places. Second, the control recovers no such association, so the estimator does not manufacture it. Third, the control places the sampling dispersion of $\hat{\alpha}$ at 0.03 (40 replicates, samples of 800–4,000 points, true exponent fixed at 2.0), against an observed standard deviation of 0.154 across the 15 national fits: the year-to-year movement in α is real, roughly five times the sampling noise in standard deviation, even though it remains small next to the two-order-of-magnitude variation in the case counts themselves. The sizes of the actual national cross-sections are not recorded in the results database, so the 8–4,000 range is chosen to bracket them rather than estimated from them.

Because $x_{\min}$ is measured in cases, it tracks the magnitude of the season, and an exponent that co-varies with the magnitude of the season is to that extent a re-encoding of it. This is the reason the ablation of Section 2.10 is the decisive test of the scaling block rather than an optional addition to the descriptive result.

3.4 The fitted law as a risk model

Equation (3) makes the descriptive result operational. Figure 9 confronts the exceedance curve implied by each national fit with the empirical exceedance frequency of the same cross-section: the fitted pair $(\alpha, x_{\min})$ reproduces the observed frequency of large seasons over roughly two decades of epidemic size, and it does so with a two-parameter summary of a cross-section of thousands of municipalities.

This is the sense in which the description is predictive before any machine learning is involved: α is not a summary of central tendency but a summary of how much of the risk sits in the tail, and it is available for every municipality even when that municipality has never experienced an extreme season.

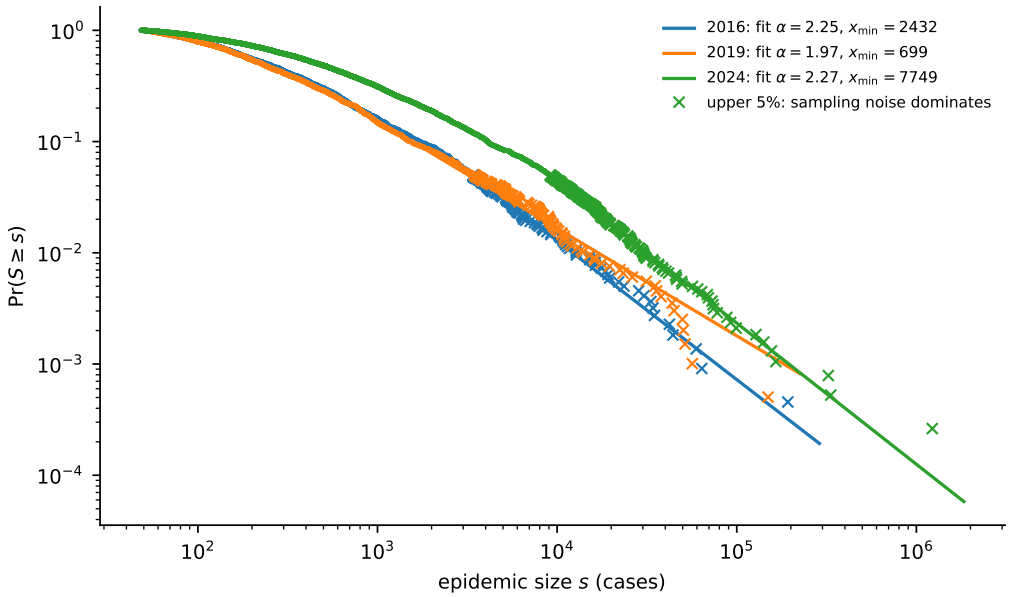


Figure 9: The fitted power law as a probabilistic forecast of exceedance. For each national cross-section, curves give $\Pr(S \geq s) = \zeta(\alpha, s)/\zeta(\alpha, x_{\min})$ evaluated at the fitted $(\alpha, x_{\min})$, and points give the empirical exceedance frequency of the same cross-section. Agreement across two decades of epidemic size is what licenses the use of $(\alpha, x_{\min})$ as a compact statement about a municipality's risk of an extreme season, and therefore as a candidate predictor in Section 3.5. Where the curves separate from the points in the extreme upper tail, the sample is too small to distinguish model error from sampling noise; those regions are marked.

3.5 Predictive skill

Table 6 reports the ablation ladder over the five forecast origins 2021–2025 at state level (26 units, 23–25 evaluated per origin). Three results organise the remainder of the paper.

The scaling block adds no predictive information. The M3→M2 contrast is null for every target under the rule fixed in Section 2.10: the block improves on the preceding rung with a 95% bootstrap interval excluding zero in 0 of 5 origins for size, 0 of 5 for duration and 1 of 5 for peak week, against the 3 of 5 required. For epidemic size the median relative change in pinball loss at $\tau = 0.5$ is -1.5% — weakly harmful — and the five per-origin differentials carry no consistent sign (two favour M3, three favour M2). The same ordering holds on every other case-scale loss: on MAE at $\tau = 0.5$ the median across origins is 1.18×10^5 cases for M2 against 1.20×10^5 for M3, and the RMSE is 2.86×10^5 against 2.95×10^5.

It is worst exactly where it was supposed to be best. The argument for α_{t-1} is that it encodes how much of the risk sits in the tail, so any gain should concentrate at high τ; Figure 11b was drawn to test precisely that. At $\tau = 0.95$ the point estimate of the M3−M2 differential is negative in *all five* origins — -24.9%, -23.7%, -0.2%, -1.6% and -18.3% of the M2 loss, median -18.3% — although with 23–25 units per origin no individual interval excludes zero. Adding $\hat{\alpha}_{t-1}$ and $\hat{x}_{\min,t-1}$ makes the upper quantile of the size forecast worse. This is what the coupling of Section 3.3 predicts: both statistics are themselves functions of the magnitude of the previous season, so at the upper tail the model is handed a second and noisier copy of S_{t-1} and weights it as if it were independent information.

The model's only real skill is in duration. Against calibrated persistence, the lagged-epidemiological rung M1 is useful for duration (median skill 0.27, positive in 4 of 5 origins, 3 of 5 with an interval excluding zero: supported) and carries almost nothing for size (median 0.10, positive in 3 of 5, 2 of 5 excluding zero: not supported). Climate (M2→M1) adds nothing on any target (0 of 5, 1 of 5 and 0 of 5) and is actively harmful for peak week in two origins, where skill falls to -0.69 in 2023 against -0.16 for M1. Peak week is essentially unpredictable at this level of aggregation under every rung. A cell-level diagnostic supports the reading that these are absences of effect rather than absences of power: over the 19×5 cells of the contrast grid, the interval excludes zero in 37% of cells for M1−M0 on size and in only 15% for M3−M2, so the design does detect effects of the size the ladder was built to find.

This inverts an impression created by an earlier full-pipeline run, which reported an out-of-bag R^2 for size of 0.345, 0.348 and 0.374 for 2022, 2023 and 2024; we report those figures now only to explain the discrepancy. That R^2 was computed on the pooled cross-section, on the log scale on which size is fitted, and it is dominated by *between*-unit differences in mean epidemic size — which a persistence forecast reproduces trivially, because the ordering of states by typical size is stable from year to year. It is not a statement about year-on-year change, which is what a forecast has to deliver. Once the loss is evaluated on the case scale against a persistence benchmark, the apparent skill evaporates. The lesson is the one this paper keeps repeating: the number depends on the benchmark, on the scale and on the comparison set, and a high R^2 against no benchmark is not evidence of forecastability.

Two origins dominate the size column and need separate mention. The 2024 season was the largest on record and 2025 fell back sharply; persistence extrapolated the record into 2025, and in absolute terms lost there much of what it had gained in 2024. The pinball losses for size are accordingly an order of magnitude larger in those two origins (1.5–1.7×10^5 cases) than in the other three (1.2–3.2×10^4), so a pooled mean over origins

would mostly describe how each rung behaves across a regime shift. We therefore report the median across origins and the sign of the per-origin differential, not a pooled average.

Figure 10 shows the out-of-sample forecasts in the form in which their errors are interpretable, and Figure 11 the ladder itself.

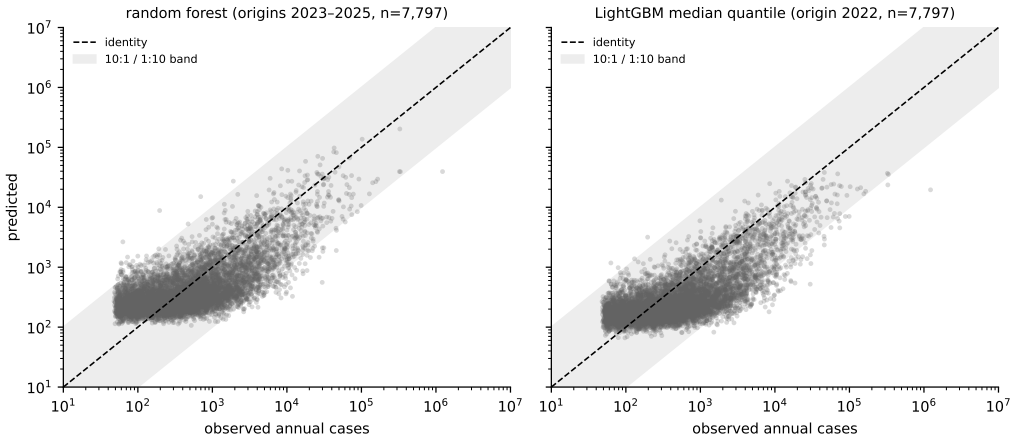


Figure 10: Out-of-sample forecasts of epidemic size. Predicted against observed annual cases on log–log axes, one point per unit and forecast origin, for the random forest (left) and the median of the gradient-boosted quantile ensemble (right); the identity line is dashed and the 10:1 and 1:10 bands are shaded. Points inside the bands are forecasts within an order of magnitude of the truth. Because $\alpha \approx 2$, a handful of points in the upper right dominate any squared-error summary, which is why the primary loss reported in Table 6 is the pinball loss rather than the RMSE.

3.6 Forecasting the tail

The classification task asks a narrower question than the regression: not how large the season will be, but whether it will fall in the upper quartile of the unit's own history. Under Equation (3) that is a question about the tail probability directly, which makes it the natural place to look for value in the scaling descriptors. Figures 12 and 13 report discrimination, calibration and attribution.

This rung was not part of the ablation pass reported in Table 6: the classification run requires its own flag in the runner and was executed separately. What follows therefore describes the fitted classifier and its attribution, and claims no ablation verdict for the scaling block on the classification task. Since the exceedance formulation is the most natural use of $(\alpha, x_{\min})$, and since the regression ladder returned a null for the same block (Section 4.7), the classification row is the one place where a positive result would still be compatible with the descriptive half of the paper. It is reported as an open question, not as support.

3.7 Error structure and spatial output

Accuracy pooled over units is not the same as accuracy where decisions are made, so we report the error structure (Figure 14) and the map a surveillance service would actually read (Figure 15).

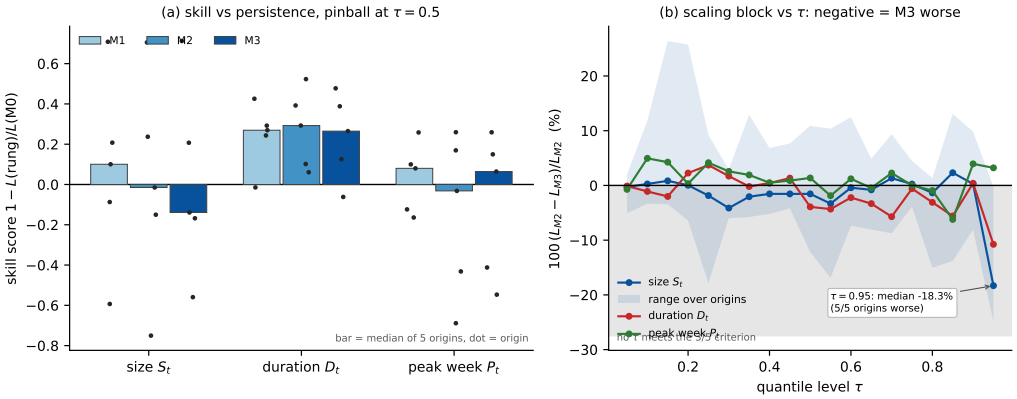


Figure 11: The ablation ladder, drawn directly from `ablation_results.csv` rather than recomputed. (a) Skill score relative to the calibrated persistence benchmark (M0) for epidemic size, duration and peak week at the primary loss (pinball, $\tau = 0.5$); bar height is the median over the five forecast origins and each dot is a single origin. The M2→M3 step isolates the contribution of the scaling descriptors $(\alpha_{t-1}, x_{\min,t-1}, \dot{\alpha}_{t-1})$. (b) The quantity the decision rule turns on, $100\,(L_{M2} - L_{M3})/L_{M2}$, against τ: a gain confined to high τ would mean the scaling information concerns extreme seasons specifically, which is what a heavy-tail description should deliver, so negative values here are the opposite of the intended effect. Lines are medians over origins and the band is the range across origins for epidemic size. The block degrades the extreme quantile of the size forecast, with a median loss of 18.3% at $\tau = 0.95$ and a negative differential in all five origins, and no τ on any target reaches the 3/5 criterion of Section 2.10. Skill scores carry no interval of their own: the unit-bootstrap interval in this analysis belongs to the paired differential, which is what Table 6 reports.

Table 6: Ablation ladder over the five forecast origins 2021–2025, state level (26 units, 23–25 evaluated per origin). Cells give the median across origins of the skill score $1 - L(\text{rung})/L(\text{M0})$ against calibrated persistence, with the range over origins in brackets; the primary loss is the pinball loss at $\tau = 0.5$ for all three targets. The last column is the M3−M2 differential at that loss: median relative change, negative meaning that the scaling block worsens the forecast, and in parentheses the number of origins in which M3 improves on M2 by sign alone; in at most one of those origins does the bootstrap interval exclude zero, so the count in parentheses is not the one the decision rule scores. The decision rule of Section 2.10 grants the rung no credit unless the improvement holds in the same direction with a 95% unit-bootstrap interval excluding zero in at least 3 of the 5 origins; no contrast involving the scaling block reaches that threshold, and no rung at all beats persistence for epidemic size. The large-season classifier is not tabulated: it was not part of this run (Section 4.7).

Target	M1	M2	M3	M3−M2
Size S_t	0.10 [−0.59, 0.71]	−0.01 [−0.75, 0.71]	−0.14 [−0.56, 0.71]	−1.5% (2/5)
Duration D_t	0.27 [−0.01, 0.43]	0.29 [0.06, 0.52]	0.26 [−0.06, 0.48]	−3.9% (1/5)
Peak week P_t	0.08 [−0.16, 0.26]	−0.03 [−0.69, 0.26]	0.06 [−0.55, 0.26]	+1.4% (3/5)

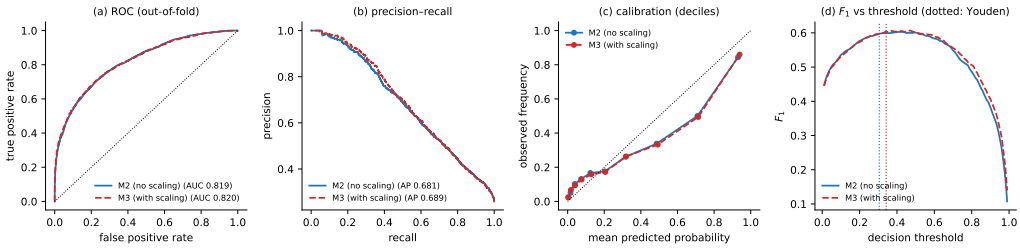


Figure 12: Classification of large epidemics. (a) ROC curves; (b) precision–recall curves, the more informative summary when the positive class is the minority; (c) calibration of the predicted probability of exceeding the 75th percentile; (d) F_1 as a function of the decision threshold, with the Youden point marked. All curves are drawn from out-of-fold predictions so that they are honest. Solid lines denote M2 (without scaling descriptors) and dashed lines M3 (with them). The pattern of interest is a calibration curve for M3 that tracks the diagonal where M2 over- or under-predicts the upper decile.

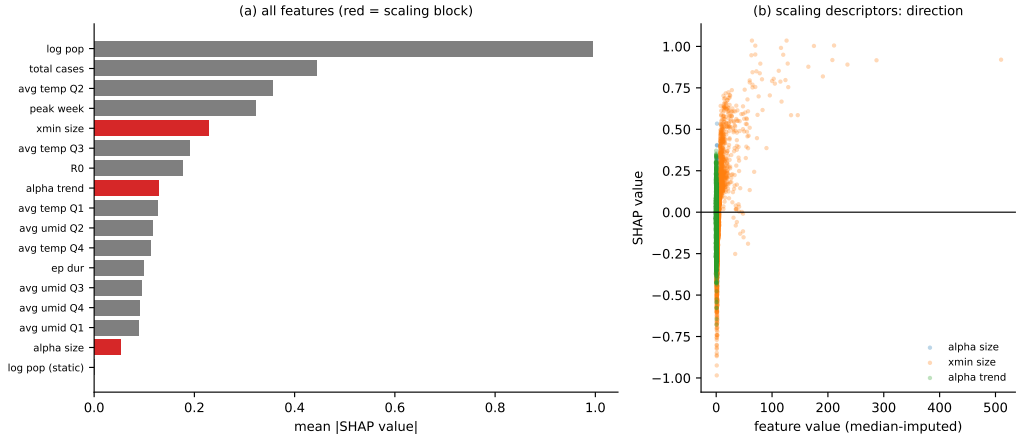


Figure 13: Where the scaling descriptors rank. Mean absolute SHAP values for the large-season classifier, pooled over forecast origins, with the power-law block highlighted. The left panel ranks all features; the right panel isolates α_{t-1}, $x_{\min,t-1}$ and $\dot{\alpha}_{t-1}$ and shows the direction of their effect on the predicted probability. A positive, monotone relation between α_{t-1} and the predicted probability would indicate that flatter-tailed histories precede larger seasons.

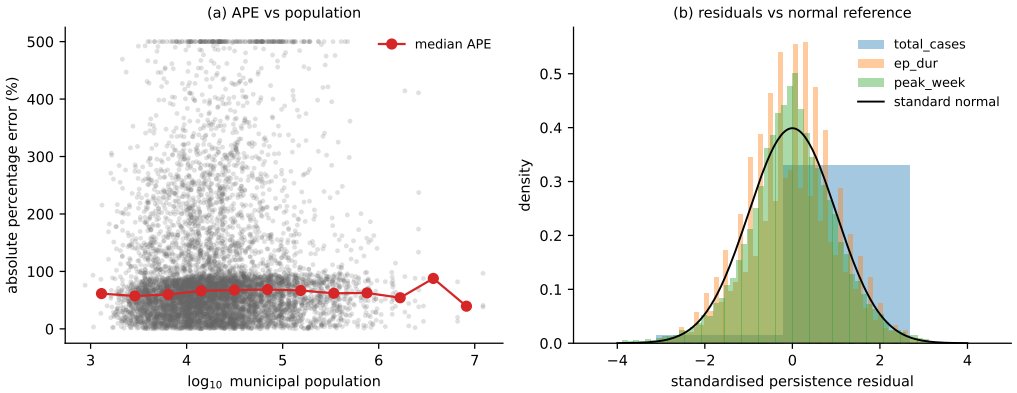


Figure 14: Error structure. (a) Absolute percentage error against municipal population, with the population filter indicated; (b) distribution of standardised residuals by target against a standard normal reference. Panel (a) documents that small municipalities are intrinsically harder to forecast, so pooled accuracy figures must be read together with the population composition of the evaluation set — a point that also applies to the published forecast comparisons this work is meant to inform.

4 Discussion

4.1 The verdict belongs to the construction

One estimator, one body of surveillance data, seven ways of cutting it, and verdicts ranging from unanimous support for a power law (L6, 15 of 15 years) to unanimous rejection in favour of an exponential (L7). Median exponents span 1.95 to 2.95 across the levels, and the selected lower bounds run from a single case to about 1.5×10^5, five orders of magnitude. Nothing distinguished the analyses except the construction: there was no second data source, no rival estimator and no software difference to which the disagreement could be attributed.

The sharpest form of the argument is the L1/L2 pair, which differ in nothing but the length of the window applied to the same weekly counts. Pooled over a decade the power law is preferred in 75% of municipalities; cut into single seasons the same counts prefer it in 23%, while significant support for the exponential rises from 4% to 21%. Between those two numbers not a single transmission parameter changes. An analysis that reports the first without the second has reported a property of its window.

We do not read this as showing that the published exponents are erroneous. The individual estimates may each be correct arithmetic. What does not survive is the inference from an exponent to a regime: the step that treats α as a property of the pathogen or the place, when it is a property of the tuple (unit, window, variable, comparison model).

4.2 Size and duration are different random variables

The cross-sectional levels settle a second, quieter disagreement. Epidemic *size* and epidemic *duration* are computed from the same municipal episodes, and they give opposite verdicts under the same estimator. Size is power-law with $\alpha \approx 2.0$ at national level and 1.97 at state level. Duration is not: the power law is rejected in all 15 national fits and in 95% of the 330 state-year fits, and the estimated exponent collapses onto a narrow

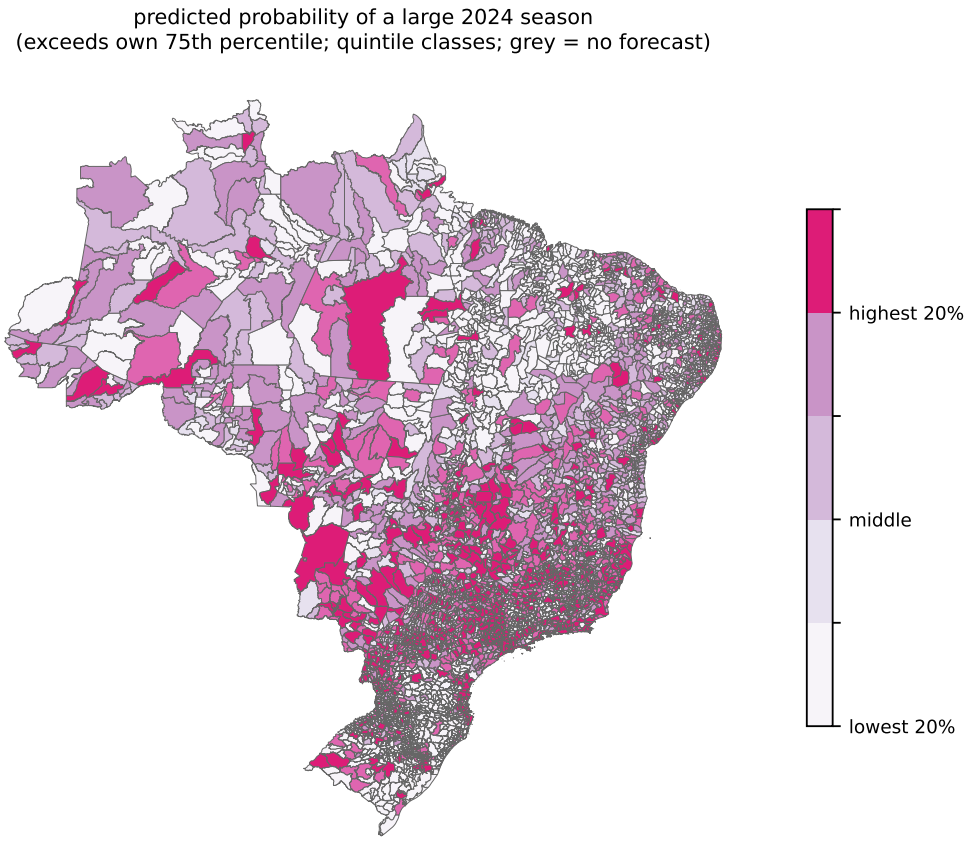


Figure 15: Probabilities of a large epidemic for a single forecast origin. Predicted probability that each municipality exceeds its historical 75th percentile of annual cases, classed in quintiles; grey denotes units without a forecast. The map is included because the calibration of the upper quintiles is a public-health claim rather than a statistical one: it is the object a surveillance service would act on, and it is falsifiable season by season.

band (2.88–2.99, SD 0.03) that presses against the estimator's ceiling of 3 in 11 of the 330 fits. That is what a bounded or exponential body returns when forced through a power-law estimator. A near-constant fitted exponent across independent cross-sections is not evidence of universality; it is evidence that the model is being asked the wrong question.

Because both variables are routinely described as "epidemic size" or "outbreak magnitude" in the literature, part of the apparent contradiction between published dengue exponents is plausibly the mixing of these two distributions. The relevant distinction is not biological before it is mathematical: a quantity bounded below by one week and above by the length of a season cannot have the support that a scale-free law requires.

4.3 What α is worth without its tuple

The practical consequence is a reporting convention, and it is cheap to adopt. An exponent should travel with the lower bound that produced it, the unit of analysis, the number of observations, and the alternative model it was tested against. Of these, $x_{\min}$ is the most informative and the most frequently omitted, because it separates two analyses that look identical in a table of results. At $x_{\min} = 2$ the fitted power law describes counts of one and two cases and the heavy-tail language is vacuous; at $x_{\min} = 762$ it describes the upper decade of a distribution and the language is warranted. The L1 and L3 constructions illustrate the point in a single line: their median exponents are almost the same (1.95 and 2.09, a difference of 0.14) while their median lower bounds differ by a factor of 170 (2 versus 341 cases). Two analyses that would be reported as agreeing on the exponent are not describing the same object.

The diagnostic is easy to apply and hard to game, since $x_{\min}$ is chosen by the same goodness-of-fit criterion that selects the model and cannot be tuned to produce a desired α. We would go further: an α reported without $x_{\min}$ should be treated as an incomplete statistic, in the same way that a correlation is incomplete without its sample size.

4.4 Why the coupling matters for prediction

Two properties of the fitted pair must be separated, because the forecasting argument depends on the first and is threatened by the second.

The first is that α is low-variance relative to its target. Across the 15 national cross-sections it ranges from 1.78 to 2.27, while annual case counts over the same years vary by two orders of magnitude. On simulated samples of 800–4,000 points with the true exponent fixed at 2.0, the sampling dispersion of $\hat{\alpha}$ is about 0.03 (Section 3.3); the observed dispersion is roughly five times larger, so the year-to-year variation is real rather than noise, but it remains small on the scale on which the target varies. That is the specific reason a shape parameter can be a candidate predictor in a setting where the observables are volatile.

The second is that α is not independent of $x_{\min}$. At the national cross-section the rank correlation between the two is 0.87 ($n = 15$, $p < 0.001$). Within states it is 0.65 at the median across the 24 states with at least six years of data, positive in 88% of them and significantly so in 67%, whereas between states it is only 0.22. The coupling is therefore a within-unit relationship over time, not a contrast between places.

This is not an artefact of the estimator. Repeating the fit on simulated data drawn from a power law whose true exponent is held fixed at 2.0, while the true lower bound is varied across the same range, recovers no positive coupling (Pearson $r = -0.20$, $n = 40$)

and an essentially unbiased exponent (mean 2.012). The correlation in the surveillance data is a property of the data, not of the method.

The consequence for prediction is direct and uncomfortable. Because $x_{\min}$ is measured in cases, it tracks the magnitude of the season: years in which municipalities produce large epidemics place the cut higher. An exponent that co-varies with the scale of the season is, to that extent, a re-encoding of the scale of the season — and the scale of the previous season is exactly S_{t-1}, which is already present in the M2 rung. The descriptive case for carrying $(\alpha, x_{\min})$ into a forecasting model is therefore weakened by the same analysis that supports it: the block may be redundant with the lagged counts it is meant to improve upon. This is a testable expectation, not an objection, and it is the expectation with which the ablation ladder was designed. If the scaling block earns its place, it must do so on the primary loss and against a model that already knows how large last season was.

4.5 What the descriptive result does license

The exceedance curve, however, is not affected by that caveat, and it is the part of the description that carries practical content. Equation (3) turns $(\alpha, x_{\min})$ into a probability for every epidemic size, and the 15 national fits imply that between 5.4% and 16.5% of municipalities exceed ten times $x_{\min}$ in a given year, and between 0.3% and 2.7% exceed a hundred times it. Those are statements a surveillance service can act on and that a subsequent season can falsify, and they require no fitted case count, only the current cross-section.

It is worth being explicit about what follows, and what does not, from $\alpha \approx 2$. The familiar inference — that a divergent second moment makes extreme seasons unpredictable — is too strong. It is a statement about second moments and about squared-error loss, not about information, and an exponent remains a scaling statement even where the variance diverges: branching-process analysis shows that a power-law-tailed offspring distribution yields a finite-time scaling law governed by the exponent alone (Falcó and Corral, 2022). A power law with $\alpha = 2$ assigns a definite probability to every exceedance level; what it denies is that the *magnitude* of the largest event can be estimated from the variance. Prediction of the tail event and prediction of its size are different problems, and proper scoring rules keep them separate. The literature that reads $\alpha \approx 2$ as a verdict on forecastability conflates the two.

4.6 Relation to existing forecast systems

Operational dengue forecasting in Brazil has largely been built on climate and population susceptibility, with machine-learning models predicting municipal epidemics months ahead from weather and immunity proxies (McGough et al., 2021), and with nowcasting platforms supplying the adjusted case series used here (Araujo et al., 2025). Those approaches ask what covariates precede an outbreak. The question posed here is orthogonal: whether the *shape* of the size distribution carries information that the covariates do not. The two are complementary rather than competing, and the ablation design is deliberately constructed so that the scaling block is judged against a model that already contains the climate and lagged-metric blocks.

There is also a methodological debt to the forecasting literature that the power-law literature has not settled. Skill has to be measured with proper scoring rules (Gneiting and Raftery, 2007) on paired loss differentials (Diebold and Mariano, 1995), not with the correlation of predicted and observed point forecasts. With $\alpha \approx 2$ the latter is dominated by the few largest outbreaks and can be moved from excellent to poor by a single season.

4.7 The predictive test

The ablation returns a negative answer, and it is worth being precise about what it rules out. The block $(\hat{\alpha}_{t-1}, \hat{x}_{\min,t-1}, \dot{\alpha}_{t-1})$ does not improve on the covariates it is added to, on any of the three targets, under a rule fixed before the runs existed; and where it has an effect at all, that effect is at the extreme upper quantile and it is a loss rather than a gain. What this does *not* say is that α is uninformative. The exponent remains a low-dimensional description of the exceedance curve that reproduces the observed frequency of extreme seasons (Section 3.4). It says that whatever information the exponent carries is already contained in S_{t-1}, and reaches the model there in a form that is easier to use.

The direction of the failure is what makes that reading credible. A feature block that is merely irrelevant should be close to ignored by a boosted tree: the split search does not select it, and the loss neither improves nor degrades much. What we observe at $\tau = 0.95$ is degradation in five origins out of five, between 0.2% and 25% of the M2 loss. That is the signature of a variable which is *selected* and then misused — what a noisy proxy for a quantity already in the model produces. The tree cannot tell $\hat{\alpha}_{t-1}$ from a second, genuine measurement of last season's magnitude, so it splits on it, and at the upper quantile, where the fit is least constrained by data, the error compounds. The coupling measured in Section 3.3 is therefore not merely a caveat about correlated predictors; it predicts the sign of the observed degradation, and the prediction is met without adjustment.

The honest summary of the predictive half is narrower than the section title suggests. The scaling description is not *forecast-useful* against a benchmark that already knows the previous season, and no amount of reporting of α will make it so, because the redundancy is structural rather than a matter of model choice or hyperparameter. It is *decision-useful* in another sense: it converts a cross-section of one season into an exceedance curve for the next, which is what a service needs in order to set a threshold, and it does so without requiring a fitted case count for the unit in question. Those two claims have been run together in the literature; separating them is the substantive contribution here.

The negative result is also a statement about the benchmark, and that should not be buried. Calibrated persistence is a strong competitor for epidemic size at state level: annual counts are dominated by a slowly varying unit-level mean, and the year-to-year innovation is small by comparison with the between-unit spread. Any model that reports only a cross-sectional R^2 — as the earlier run of this pipeline did — will appear successful while adding nothing over the last observation. We would put the point more strongly: for a target with this autocorrelation structure, a study that reports forecast skill without a persistence benchmark has not reported forecast skill.

Two refinements keep the conclusion from overreaching. First, α becoming uninformative *as a lagged single-season estimate* is not the same as the exceedance description being uninformative, and the two were not tested by the same instrument: the ladder tests the former, and the reproduction of the observed frequency of extreme seasons tests the latter. Second, persistence being hard to beat is not a defect of the models but a property of the target. For epidemic size the predictable component is largely the unit's own level, which is already known; what is left is the innovation, and no covariate in the panel – climate included, at state level – captured it.

Three qualifications bound all of this. The panel is at state level, so 26 units and 23–25 per origin; the bootstrap intervals are correspondingly wide, and the *absence* of a significant M3−M2 interval is weaker evidence than its presence would have been. The classification rung — the task for which the exceedance formulation is the most natural use of the fitted pair — was not part of these runs, so the negative result applies to the quantile regressions and is silent about the classifier. And the block was tested in the form

the pipeline produces it: an exponent estimated on a single season, whose $x_{\min}$ therefore tracks that season's magnitude. A block built to avoid the coupling — an exponent estimated over a rolling multi-year window that excludes the target year, with $x_{\min}$ held fixed across years so that it is a property of the unit rather than of the season — is a different feature set, and nothing here speaks to it. That is the version worth testing next, and the ladder as built would test it without modification.

4.8 Limitations

Several limitations bound these conclusions.

The case series are nowcast-adjusted. The variable analysed is the estimated case count produced by the platform's nowcasting model, not the raw notification count. This is the right series for a retrospective description, because raw notifications are incomplete in recent weeks, but it means that the upper tail inherits whatever smoothing the nowcast applies. If the nowcast shrinks extreme values, the fitted exponents are biased towards the bulk, and the size distribution reported here is somewhat less heavy than the notification distribution would be.

The state level is an arithmetic aggregate. State-level duration is the mean of municipal durations and state peak week is the median of municipal peaks. Neither is the duration or the peak of the aggregate epidemic, which is the quantity a state health secretariat actually experiences. The state-level results should be read as properties of the municipality distribution within a state, not as properties of the state epidemic.

The cross-sections are short. The national levels rest on 15 annual fits and the state levels on three to 15 years per state, so the correlations reported in Section 3.3 are estimated on small samples even where the underlying cross-sections contain thousands of municipalities. The within-state estimates are more robust than the national ones because they pool over 24 states.

The trees cannot extrapolate. This limitation cuts across both halves of the paper and deserves emphasis. Tree ensembles predict within the support of their training responses. The season of greatest public-health interest is the exceptional one, and under $\alpha \approx 2$ the largest season a municipality will experience is systematically outside the support of its observed history. The models are therefore weakest precisely where the heavy tail places the risk, and no amount of feature engineering removes that: it is a property of the estimator class. It is also the reason the quantile models are reported alongside the point forecasts, since a quantile regression at $\tau = 0.95$ at least attempts to represent the upper region rather than interpolating within the observed range.

The classification threshold is a choice. Exceeding the 75th percentile of a unit's own history defines a large season; at the 90th percentile the positive class becomes small enough that the balanced accuracy of any classifier falls, and the comparison between rungs loses power. The threshold is therefore a design parameter, fixed before the runs and reported with them.

The classification rung was not evaluated here. The ablation in Table 6 covers the three quantile regressions. The rung that adds the scaling descriptors to the large-season classifier requires a separate flag in the runner and was not part of the run that produced these numbers, so the negative result of Section 4.7 applies to the regressions and is silent about the classifier. Since the exceedance formulation is the most natural use of $(\alpha, x_{\min})$, this is the gap most worth closing, and closing it means re-running the same script with the classifier enabled.

The exponent cannot exceed three. The fitting routine bounds α above by 3, so a tail steeper than that is reported as 3 rather than as its unconstrained maximum. For the size

results this is immaterial, since no national size fit exceeds 2.27; for durations it is not, and 11 of the 330 state-year fits sit at the bound. This does not weaken the conclusion that durations are not power-law distributed — a bound that the data press against is itself evidence against the fitted family — but the numerical value of those particular exponents should not be read as an estimate.

Predictors are ecological and coarse. No serotype composition, no seroprevalence and no vector index enters the model, although all three are plausible determinants of season size. To the extent that these are stable within a unit, they are absorbed by unit-level effects; to the extent that they vary by year, they are unmodelled heterogeneity that the scaling block may be inadvertently proxying.

4.9 Implications for surveillance practice

Three practices follow, none of them expensive.

First, report the tuple. An exponent accompanied by its unit, its $x_{\min}$, its sample size and its comparison model can be assessed by a reader; an exponent alone cannot, and the seven levels here show that the same exponent can mean opposite things.

Second, use $x_{\min}$ as a screening diagnostic before interpreting any heavy tail, and treat $x_{\min}$ within the bulk of the data as a signal that the question needs restating rather than a small technical detail.

Third, do not translate a heavy tail into a claim of unpredictability without specifying the loss. A distribution without a finite variance constrains squared-error point forecasts and leaves probabilistic exceedance statements intact — and the exceedance statement is usually the one a health service needs, since it is the basis for deciding how many beds to hold in reserve.

5 Conclusion

The distribution of dengue epidemic size in Brazil is heavy-tailed with an exponent close to two, and this is one of the few robust facts in the applied literature on epidemic scaling. The distribution of epidemic duration is not heavy-tailed at all. Which of these statements a study reports, and whether it reports a power law or an exponential, is determined by the construction it imposes on the data: seven levels applied to one dataset produced median exponents between 1.95 and 2.95, unanimous support for a power law at one level and unanimous rejection at another, with a single change of temporal window sufficient to move the verdict from 75% to 23% support. That heterogeneity is methodological before it is biological, and reporting α without its lower bound, its unit, its sample size and its comparison model makes the disagreement unresolvable.

Two consequences follow for forecasting, and the ablation settles the second. The encouraging one is that $(\alpha, x_{\min})$ fixes the entire exceedance curve of a municipality, and the national fits reproduce the observed frequency of extreme seasons over two decades of epidemic size using two parameters. The cautionary one was that the estimated exponent co-varies with the scale of the season within units, so a model carrying α_{t-1} might be re-encoding S_{t-1}. The ladder was built to answer that question and the answer is that it is. Against a benchmark that already knows the previous season, the scaling block fails the pre-registered test on size, on duration and on peak week — 0, 0 and 1 of the 5 origins against the 3 required — and at the 95th percentile of the size forecast it degrades the loss in every origin tested, by up to a quarter of the benchmark loss. The exponent describes the tail; it does not help to forecast it once the last observation is in the model.

What generalises beyond this dataset is the form the question must take. The claim that a scaling description improves forecasts is only meaningful against a specified benchmark, on a proper scoring rule, out of sample, with the scaling block isolated from the blocks it is added to, and with every leakage channel closed — including the validation split, which fails silently. And the claim that a heavy tail makes epidemics unpredictable should be retired in favour of the narrower one it actually supports: that the size of the largest event is not estimable from the second moment, while its probability is.

A Verification of reported quantities

Every quantity in Table 4 is obtained from `powerlaw_results.db` by the queries in `Manuscript/verification/`. The comparison columns are computed as

`R>0` ⇒ PL preferred; `R>0 AND p<0.05` ⇒ PL supported; `R<0 AND p<0.05` ⇒ exponential supported.

Note that `year = 0` in the `episcanner` tables is a pooled all-years fit, not a calendar year, and is excluded from all year counts; the national levels therefore have 15 annual fits, not 16.

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